Level of Digital Literacies Among Austrian College Students Assessed with an Online Survey

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Abstract. The digital age offers opportunities for the development of online information and communications technology (ICT) systems that emphasize the important role of digital literacy, including communication between people and the values necessary for working in an online environment, which greatly influences the processes and practices of learning and teaching. Especially in the field of mobile learning, digital and data-evaluating skills are of great importance, as this type of technologically extended learning is based on a comprehensive use of cloud and internet-based services.

The purpose of this study was to evaluate the German version of a validated 54-item questionnaire and to assess the level of digital literacies and digital readiness of undergraduate and graduate students at the Carinthia University of Applied Sciences.

An online questionnaire was developed based on the items designed by Kurtz and Peled [1]. The questionnaire included 54 items divided into seven areas: data research and retrieval; data validation; data management; data processing; teamwork; integrity awareness; and social responsibility. The items were translated into German language and their validity was evaluated with Cronbach's Alpha in a pre-study on 155 students. After improvement of some items, all students from the Carinthia University of Applied Sciences were asked to complete the questionnaire.

The first key outcome of this study is the validation of the German version of a validated 54-item questionnaire. The second key outcome is the level of digital literacies and digital readiness of undergraduate and graduate students at the Carinthia University of Applied Sciences. Finally, we want to find out if there are differences in the digital competencies between the various degree programs of the Carinthia University of Applied Sciences.

Our findings should enable faculty and educational policymakers to identify the strengths and weaknesses of digital literacies of students from the Carinthia University of Applied Sciences.

Keywords: Digital learning literacies · 21st century skills · Learning skills

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1 Introduction

1.1 Digital Literacies

Models of online and blended learning are gaining popularity in higher education, as institutions seek to support and facilitate modern, flexible educational opportunities for students, and create interactive digital learning experiences that are engaging and effective in undergraduate and postgraduate education. In part, the challenge of providing an interactive learning experience for students in large class groups and concerns about the quality of pedagogy in this environment have been key catalysts for the reconsideration of teaching approaches in higher education, as well as to overcome issues of distance between learners and instructors, and to facilitate increased peer interaction and collaboration. For information and digital literacy skills development, in particular, the adoption of online modes is increasingly popular [2].

The rationale for formulating '21st century skills' is that the 21st century is very different from the 20th century in terms of the skills required for work, citizenship, and self-accomplishment [3]. These developments call for a new form of learning, which should be receptive to this continually changing world. In fact, certain skills and ways of thinking are becoming more and more essential for college students [4].

Part of the set of skills required for the 21st century are digital literacies, which might best be defined as "those capabilities which fit an individual for living, learning, and working in a digital society" [5]. In addition, digital literacy helps an individual navigate current digital environments actively, collaboratively, and in a participative manner [1].

Digital literacy includes communication between people and the tools necessary for working in an online environment, which greatly influences the processes and practices of learning and teaching [6]. These processes influence the training and professional development of college lecturers that are aware of the requirements of the digital age and its implications for education.

1.2 Conceptual Framework

The idea of this project was to apply a validated questionnaire on digital literacy to Austrian college students. The original questionnaire was conceived in English language and needed translation into German first.

Seven domains represent digital literacy [1]: Data Collection; Evaluation of Data; Data Management; Data Processing; Teamwork; Integrity awareness; and Social responsibility. These domains are the basis for one's ability to manage in the complex, ever-evolving 21st century environment. This capability is described as digital readiness, which is an interpretation of digital literacy, and is defined as the extent to which an individual or group has the knowledge and ability to integrate cognitive skills appropriate for activity in digital environments [7, 8].

1.3 Aim of the Study

The aim of this study was to implement a German version of a validated questionnaire of digital literacy of college students. Precisely, we wanted to answer the following research questions:

1) Is the internal consistency of the German version comparable to the internal consistency of the original version?
2) Are there any gender-specific differences in the seven dimensions of digital literacy?
3) Are there any differences between bachelor and master students in the seven dimensions of digital literacy?
4) Are there any differences between full-time and extra-occupational students in the seven dimensions of digital literacy?

2 Methods

2.1 Questionnaire

This online questionnaire was based on a validated 54-item questionnaire by Kurtz and Peled [1] and translated into German. The following Likert-scale for answering the items was applied: 1) Strongly disagree; 2) Somewhat disagree; 3) Neither disagree nor agree; 4) Somewhat agree; 5) Strongly agree.

In addition, the following demographic and study-specific variables were assessed: age, gender, field of study and whether it was a full-time study or extra-occupational education.

The online-questionnaire was designed and distributed using the QuestorPro (Blubbsoft GmbH, Berlin, Germany). All students of the Carinthia University of Applied Sciences (n = 2.167) received an e-mail with the invitation to participate in this survey (Tables 1, 2, 3, 4, 5, 6 and 7).

Table 1. Translation of items for the scale "Social Responsibility" into German.

Item	Original	German translation
Item 1	I adhere to the rules of discourse and proper behavior in social networks	Ich halte mich an die Regeln des Diskurses und des richtigen Verhaltens in sozialen Netzwerken
Item 2	I make sure not to reveal information about organizations without consent	Ich stelle sicher, dass Informationen über Organisationen nicht ohne Einwilligung veröffentlicht werden
Item 3	I make sure not to hurt others – people and organizations – online	Ich stelle sicher, dass ich andere - Menschen und Organisationen - nicht online verletze

Table 2. Translation of items for the scale "Data Collection" into German.

Item	Original	German translation
Item 1	I know when I need to look for information	Ich weiß, wann ich nach Informationen suchen sollte
Item 2	I am able to identify information for research	Ich kann Information zu Forschungszwecken richtig identifizieren
Item 3	I am able to collect information from the web	Ich kann Informationen im Internet sammeln.
Item 4	I can define the objective of the search	Ich kann das Suchobjekt genau definieren
Item 5	I can articulate what information I need	Ich kann formulieren, welche Informationen ich brauche
Item 6	I know how to search effectively	Ich weiß, wie man effektiv sucht
Item 7	I can define research terms	Ich kann Forschungsbegriffe definieren.
Item 8	I can distinguish between types of search	Ich kann zwischen Sucharten unterscheiden
Item 9	I can retrieve information from various sources	Ich kann Informationen aus verschiedenen Quellen abrufen
Item 10	I am able to collect information from databases	Ich kann Informationen aus Datenbanken sammeln
Item 11	I am able to re-locate information	Ich kann Informationen wiederfinden
Item 12	I can re-locate a specific web page	Ich kann eine bestimmte Webseite wiederfinden

Table 3. Translation of items for the scale "Evaluation of Data" into German.

Item	Original	German translation
Item 1	I am able to judge the degree to which information is practical or satisfies the needs of the task	Ich bin in der Lage zu beurteilen, inwieweit eine Informationen praktisch ist oder die Bedürfnisse meiner Aufgabe erfüllt
Item 2	I am able to determine the information required for a specific task	Ich kann bestimmen, welche Information für eine bestimmte Aufgabe notwendig ist
Item 3	I am able to assess the accuracy of information	Ich bin in der Lage, die Genauigkeit einer Information zu beurteilen
Item 4	I am able to assess the credibility of information	Ich bin in der Lage, die Glaubwürdigkeit einer Information zu beurteilen
Item 5	I am aware of the difference in credibility of information from various sources	Ich bin mir der unterschiedlichen Glaubwürdigkeit von Informationen aus unterschiedlichen Quellen bewusst

Table 4. Translation of items for the scale "Integrity Awareness" into German.

Item	Original	German translation
Item 1	I understand the ethical consequences of the use of technology	Ich weiß, welche ethischen Folgen der Einsatz von Technologie hat
Item 2	I understand the social consequences of the use of technology	Ich weiß, welche sozialen Folgen der Einsatz von Technologie hat
Item 3	I do not acquire digital information, files, programs, databases, etc. via illegal means	Ich nutze keine illegalen Mittel, um digitale Informationen, Dateien, Programme, Datenbanken usw. zu sammeln
Item 4	I do not use technology for purposes that are intimidating or threatening	Ich setze Technologie nicht ein, um jemanden einzuschüchtern oder ihm/ihr zu drohen
Item 5	I am aware of the prohibition of illegal file download	Ich weiß, dass das illegale Herunterladen von Dateien verboten ist
Item 6	I am aware of copyright issues	Ich weiß, dass es Probleme mit dem Urheberrecht geben kann
Item 7	I am aware of appropriate acknowledgement of sources I use	Ich weiß, dass ich die von mir genutzten Quellen angemessen referenzieren muss
Item 8	I am aware of the danger of being online to my data	Ich weiß, dass Internetnutzung meine Daten gefährden kann
Item 9	I am aware of cyberbullying issues	Ich bin mir der Cybermobbing-Problematik bewusst
Item 10	I am aware of identity theft issues	Ich weiß, dass es Probleme mit Identitätsdiebstahl geben kann
Item 11	I am aware of e-theft issues	Ich weiß, dass es Probleme mit Gelddiebstählen im Internet geben kann
Item 12	I am aware of the danger from my online activities	Ich weiß, dass meine Internet-Aktivitäten Gefahren mit sich bringen
Item 13	I am aware of the influence my online data has	Ich weiß, welchen Einfluss meine Online-Daten haben
Item 14	I am able to identify/avoid online fraud or identity theft situation	Ich kann erkennen, wenn im Internet ein Betrugs- oder Identitätsdiebstahl geschieht und weiß, wie ich diese Situationen vermeiden kann
Item 15	I am able to protect myself from online predators	Ich kann mich vor Internet-Gaunern schützen

Table 5. Translation of items for the scale "Teamwork" into German.

Item	Original	German translation
Item 1	During the preparation of a joint task, I know how to fit in among team members	Bei der Vorbereitung einer gemeinsamen Aufgabe weiß ich, wie ich mich in die Gruppe einfügen kann
Item 2	During the preparation of a joint task, I share my thoughts and insights with my peers	Bei der Vorbereitung einer gemeinsamen Aufgabe teile ich meine Gedanken und Erkenntnisse mit meinen Gruppenmitgliedern
Item 3	During the preparation of a joint task, I know that I have an influence on the work process	Bei der Vorbereitung einer gemeinsamen Aufgabe weiß ich, dass ich den Arbeitsprozess beeinflussen kann
Item 4	During the preparation of a joint task, I know what is expected of me	Bei der Vorbereitung einer gemeinsamen Aufgabe weiß ich, was von mir erwartet wird
Item 5	While performing a joint task I feel that my contribution to the team is meaningful	Bei der Durchführung einer gemeinsamen Aufgabe habe ich das Gefühl, einen sinnvollen Beitrag für die Gruppe zu leisten
Item 6	My peers are aware of my abilities and of what I can contribute	Meine Gruppenmitglieder sind sich meiner Fähigkeiten und meines Beitrags bewusst
Item 7	I have no reservation regarding joint tasks	Ich stehe gemeinsamen Aufgaben unbefangen gegenüber
Item 8	I like to work with my peers on a joint task	Ich arbeite gern in Gruppen an gemeinsamen Aufgaben

Table 6. Translation of items for the scale "Data Management" into German.

Item	Original	German translation
Item 1	When I store a file, I give it a specific name	Wenn ich eine Datei speichere, gebe ich ihr einen bestimmten Namen
Item 2	I story my files in designated folders	Ich speichere meine Dateien in entsprechenden Ordnern
Item 3	I tag information	Ich zeichne meine Informationen mit entsprechenden Attributen aus („Tags")

Table 7. Translation of items for the scale "Data Processing" into German.

Item	Original	German translation
Item 1	I am able to interpret information from multiple sources	Ich bin in der Lage, Informationen aus mehreren Quellen zu interpretieren
Item 2	I am able to analyze information from multiple sources	Ich bin in der Lage, Informationen aus mehreren Quellen zu analysieren
Item 3	I am able to synthesize information from multiple sources	Ich bin in der Lage, Informationen aus mehreren Quellen zusammenzuführen
Item 4	I am able to write an appropriate response to a post	Ich kann eine angemessene Antwort auf einen Beitrag („Post") verfassen
Item 5	I am able to use ICT to design or create new information from information already acquired	Ich bin in der Lage, IKT zu nutzen, um neue Informationen aus bereits gesammelten Informationen abzuleiten oder zu entwickeln
Item 6	I am able to visually organize data for learning purposes	Ich bin in der Lage, Daten zu Lernzwecken visuell aufzuarbeiten
Item 7	I can represent knowledge in a variety of ways such as PPT, website, blogs, etc.	Ich kann Inhalte auf unterschiedliche Art und Weise darstellen, wie z. B. PowerPoint, Webseiten, Blogs usw
Item 8	I am aware of the difference in written, graphic or video representations	Ich bin mir der Unterschiede zwischen Präsentationen im Text-, Bild- oder Videoformat bewusst

2.2 Statistical Analyses

Statistical analyses were conducted with IBM SPSS (version 25). As measures of central tendency and dispersion, mean values and standard deviations were calculated. The reliability of the different scales was assessed using Cronbach's alpha. Independent two-sample t-tests were applied to determine if the means of two sets of data with equal variances (approved with a Levene test) were significantly different from each other.

The significance level was set to 5%; therefore, p-values <0.05 were considered as statistically significant.

3 Results

3.1 Demographics

586 students, corresponding to a response rate of 27%, filled out the questionnaire. The majority of students was female (67.2%) and enrolled a full-time study (75.6%). The students were on average 25.4 $\pm$ 7.5 years old.

Most of the students were undergraduate (84.6%), and only 15.4% were enrolled in a master program.

3.2 Reliability of the German Version

A comparison of the values of Cronbachs's alpha between the original [9] and the German version of the questionnaire can be found in Table 8. An excellent internal consistency was found for the scale "Data Collection". The scales "Data Evaluation", "Data Processing", "Integrity Awareness" and "Teamwork" showed good internal consistencies. The internal consistency of the scale "Social Responsibility" was good. Only the scale "Data Management" showed a poor internal consistency.

Table 8. Reliability of the German questionnaire.

Scale	Cronbach's alpha of original version	Cronbach's alpha of German version
Data Collection	0.939	0.922
Data Evaluation	0.885	0.834
Data Management	0.758	0.502
Data Processing	0.917	0.866
Integrity Awareness	0.934	0.831
Social Responsibility	0.888	0.764
Teamwork	0.911	0.866

3.3 Comparative Statistics

A comparison of scales between male and female students can be found in Table 9. In six out of seven dimensions, there were no statistically significant differences between the genders. Only in "Social Responsibility", female students scored significantly higher than male students did.

Table 9. Comparison of scales between male and female students (mean values and standard deviations). Type-I error probabilities were derived from t-tests.

Scale	Females (n = 385)	Males (n = 188)	P-value
Data Collection	3.93 ± 0.64	3.95 ± 0.62	0.698
Data Evaluation	3.95 ± 0.62	4.03 ± 0.61	0.148
Data Management	4.05 ± 0.69	3.99 ± 0.75	0.311
Data Processing	3.96 ± 0.63	4.00 ± 0.58	0.429
Integrity Awareness	4.36 ± 0.45	4.3 ± 0.49	0.168
Social Responsibility	4.56 ± 0.58	4.34 ± 0.69	0.0001**
Teamwork	3.99 ± 0.68	4.05 ± 0.61	0.323

A comparison of scales between bachelor and master students can be found in Table 10. In the dimensions "Data Management", "Social Responsibility" and "Teamwork", there were no statistically significant differences between bachelor and master students. However, in the dimension "Data Collection", "Data Evaluation", "Data Processing" and "Integrity Awareness", master students scored significantly higher than bachelor students did.

Table 10. Comparison of scales between bachelor and master students (mean values and standard deviations). Type-I error probabilities were derived from t-tests.

Scale	Bachelor (n = 484)	Master (n = 88)	P-value
Data Collection	3.90 ± 0.63	4.08 ± 0.62	0.016*
Data Evaluation	3.94 ± 0.63	4.11 ± 0.6	0.021*
Data Management	4.02 ± 0.71	4.08 ± 0.74	0.482
Data Processing	3.94 ± 0.63	4.13 ± 0.59	0.010*
Integrity Awareness	4.32 ± 0.49	4.44 ± 0.39	0.037*
Social Responsibility	4.47 ± 0.63	4.57 ± 0.58	0.168
Teamwork	4.00 ± 0.67	4.07 ± 0.64	0.345

A comparison of scales between full-time and extra-occupational students can be found in Table 11. In six out of seven dimensions, there were no statistically significant differences between full-time and extra-occupational students. Only in "Integrity Awareness", extra-occupational students scored significantly higher than full-time students did.

Table 11. Comparison of scales between full-time and extra-occupational students (mean values and standard deviations). Type-I error probabilities were derived from t-tests.

Scale	Full-time	Extra-occupational	P-value
Data Collection	3.92 ± 0.63	3.96 ± 0.66	0.535
Data Evaluation	3.94 ± 0.63	4.04 ± 0.62	0.088
Data Management	4.02 ± 0.72	4.04 ± 0.72	0.832
Data Processing	3.96 ± 0.62	3.98 ± 0.65	0.786
Integrity Awareness	4.31 ± 0.51	4.41 ± 0.42	0.043*
Social Responsibility	4.48 ± 0.61	4.49 ± 0.69	0.951
Teamwork	4.04 ± 0.65	3.91 ± 0.71	0.058

4 Discussion

The aim of this study was to implement a German version of a validated questionnaire of digital literacy of college students. Digital literacy includes a variety of skills relevant to best practices in digital space. In this study, we have expanded the areas of digital literacy testing into seven areas of literacy. The findings reveal an internal consistency of the German version in all but one dimensions comparable to the internal consistency of the original version. Only the scale "Data Management" showed a poor internal consistency. These findings suggest a statistical firmness of the German questionnaire and that it can be used as valid assessment instrument of students' digital literacy.

Another aim of this study was to seek for background characteristics differences, mainly: gender, full-time vs extra-occupational and level of studies. In six out of seven dimensions, there were no statistically significant differences between the genders. Only in "Social Responsibility", female students scored significantly higher than male students did. Similarly, there were no statistically significant differences between the full time vs extra occupational students. Only in "Integrity Awareness", extra-occupational students scored significantly higher than full-time students did.

A breakdown of undergraduate and graduate students pointed to a significant difference in favor of graduate students. In the dimensions "Data Collection", "Data Evaluation", "Data Processing" and "Integrity Awareness", master students scored significantly higher than bachelor students did. These findings imply that, compared to master students, a considerable share of young bachelor students lack digital skills appropriate for the 21st century.

Lean and Jones (2011) suggested that in order to understand the changes that are taking place for learners in today's higher education more attention needs to be paid to textual practice around learning and less upon the technologies and their applications [10].

The practical significance of our findings is that some of the bachelor students are partially prepared for optimal functioning in the world of advanced technologies and the era of the "information society." The research findings emphasize the need to train teaching staff in higher education institutions to give the best preparation to their bachelor students. In addition, the findings of the study, which offer an updated rating for the examination of digital literacy types, can serve as a tool for teaching staff in institutions of higher education, as well as policymakers to develop, as necessary, the types of digital literacy required among students in institutions of higher learning.

Therefore, McGuinness and Fulton (2019) suggested that instructors should seek to strategically embed interactive digital learning objects in their courses at defined points of need in a logical structure, e.g., to reinforce classroom-based learning, or to support specific skill development. Potential disruption to learning should be minimized by following best practice guidelines to ensure ease of access, a seamless user experience, and timely feedback, as well as providing adequate support for rapid resolution of technical glitches [2].

The next step could be an m-learning based test about the digital competences of students, based on the DigComp Framework of the European Union [11].

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